

Personal information

 **Romain Claret**
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Research Interests

Neuroevolution · Bio-Inspired · Open-Endedness · Artificial Life · Robotics · Space

Education

- 2020 –  **Ph.D. Computer Science** with Prof. Dr. Kilian Stoffel and Prof. Dr. Paul Cotofrei.
Evolving Neural Networks toward Humanity-inspired Artificial Collective Intelligence.
University of Neuchâtel, Neuchâtel, Switzerland
- 2018 – 2020  **M.Sc. HES-SO Software Engineering** with Prof. Dr. Jean Hennebert.
Multi-hop Multi-turns Question-Answering Chatbot using Sub-Knowledge Graphs.
University of Applied Sciences and Arts of Western Switzerland, Lausanne, Switzerland
- 2013 – 2016  **B.Sc. HES-SO Software Engineering** with ing. info. dipl. EPF Marc Schaefer.
Anonymous and Decentralized Browser-based Data Sharing Service.
University of Applied Sciences and Arts of Western Switzerland, Neuchâtel, Switzerland

Complementary Education

- 2024 – 2025  **GECCO Summer School Computer Science Workshops / Presenting (4 days).**
RMIT University, Melbourne, Australia / University of Málaga, Málaga, Spain
- 2022 – 2024  **BENEFRI Summer School Computer Science Workshops / Presenting (6 days).**
University of Neuchâtel, Bern, and Fribourg
- 2023  **CUSO Winter School Computer Science Workshops / Presenting (28 Hours).**
University of Neuchâtel, Switzerland
- 2020 – 2024  **CUSO Doctoral Training Computer Science and Skills (112 Hours).**
University of Neuchâtel, University of Geneva, University of Lausanne, Online
- 2020  **Required Doctoral Coursework Major in Computer Science (18 ECTS).**
University of Neuchâtel (UNINE), Neuchâtel, Switzerland
- 2010 – 2012  **Undergraduate Major in Physics (5 ECTS).**
Swiss Federal Institute of Technology Lausanne (EPFL), Lausanne, Switzerland
- 2009 – 2010  **Undergraduate Major in Physics & Mathematics (35 Undergraduate Credits).**
Northeastern University, Boston, Massachusetts, United States of America
- 2009  **Extension School (4 Undergraduate Credits).**
Followed the class of Prof. Dr. Wolfgang Rueckner, Principles of Physics IT (PHYS E-1B).
Harvard University, Cambridge, Boston, Massachusetts, United States of America
- 2008  **Summer School (8 Undergraduate Credits).**
Followed the class of Dr. Jonathan Weintroub, Laboratory Electronics (PHYS S-123).
Harvard University, Cambridge, Boston, Massachusetts, United States of America

Experience

- 2023 –  **Visiting Researcher** at UCD Natural Computing Research & Applications Group. (Supervised by Prof. Michael O'Neill).
University College Dublin, Dublin, Ireland.
- 2020 –  **Doctoral Assistant** at the Information Management Institute. Applied Mathematics & Databases for 1st-year Bachelor. (Supervised by Dr. Paul Cotofrei).
University of Neuchâtel, Neuchâtel, Switzerland.

Experience (continued)

- 2017 – 2018  **IT Independent.** Consulting in Software Engineering and Blockchain. The tasks were to advise technologies, define projects, make prototypes, and conduct workshops.
Geneva area, Lausanne area, and Versicherix AG, Switzerland.
- 2016 – 2017  **Co-Founder** at a Blockchain-based InsurTech Startup. Lead in blockchain and innovation. The tasks were to make software architectures, high-level product schematics, management, workshops, prototypes, and documentation. Administration, fundraising, exhibitions, partnerships, and market studies.
Versicherix AG, Solothurn, Switzerland.
- 2010 – 2015  **Founder** at a Multimedia Streaming Startup. Lead developer and executive. The tasks were to make software architectures, implement prototypes & software, manage digital rights, build the business model, perform market studies, conduct fundraising, create partnerships with film studios, and comply with copyright laws.
Libacy, Neuchâtel, Switzerland.
- 2010  **Summer Internship** at Jenks Vestibular Physiology Lab under the supervision of Asst. Prof. Dr. Faisal Karmali. Design an experiment to identify a link between vision and the vestibular system. The tasks were to build the setup in 3D, build the setup by adapting a hydraulic flight simulator, run Matlab simulations, experiment on human subjects, and interpret the results.
Massachusetts Eye & Ear Infirmary, Harvard Medical School, Boston, United States of America
- 2012  **Summer Internship** in Watchmaking R&D using 3D computer-aided design (CAD).
Manufacture Claret, Le Locle, Switzerland
- 2006  **Summer Internship** in Watchmaking manufacturing, chamfering, and technical control.
Manufacture Claret, Le Locle, Switzerland
- 2005  **Summer Internship** in Watchmaking disassembly, reassembly and customization of a mechanical pocket watch.
Manufacture Claret, Le Locle, Switzerland
- 2004  **Summer Internship** in Watchmaking 3D computer-aided design (CAD) construction and technical drawing of watch movements.
Manufacture Claret, Le Locle, Switzerland

Teaching

- 2025  **Master's Thesis Supervisor.** Hyperparameter Optimization for Neuroevolution. Thesis defended with highest distinction.
University of Neuchâtel, Neuchâtel, Switzerland
- 2020 –  **Teaching Assistant.** Applied Mathematics (Analysis and Linear Algebra). 4 hours lecture and QA per week per semester given to first-year bachelors in Economical Science and Data Science.
University of Neuchâtel, Neuchâtel, Switzerland
-  **Teaching Assistant.** Databases (Modelization, SQL, NoSQL, Visualization). 2 hours QA per week during spring semester given to third-year bachelors in Economical Science and Data Science.
University of Neuchâtel, Neuchâtel, Switzerland
- 2020 – 2022  **Guest Lecturer.** “Demystifying Artificial Intelligence for Health Professionals”. 3 hours yearly lecture given to second-year physicians as part of their optional courses.
University of Geneva, Geneva, Switzerland
- 2021  **Teaching Assistant.** Digital Management and Processing of Multimedia Data (Analyze data, Identify and Formulate issues, Create visualizations). 4 days of project supervision and QA during the fall semester (block course) given to bachelors in Economical Science and Law.
University of Neuchâtel, Neuchâtel, Switzerland

Activities in panels, and commissions

- 2025 –  **UniNE**. Member of the University Assembly of the University of Neuchâtel.
- 2023 –  **UniNE**. Committee Member of the Mid-level staff Association (ACINE).
- 2015 – 2017  **HES-SO**. Member of the Participatory Council of the Engineering domain.

Active Memberships in Scientific Societies

Active Memberships in Scientific & Industry Societies

- 2025 –  **SRA**. Member of the Swiss Robotics Association.
- 2022 –  **IEEE**. Member of the Institute of Electrical and Electronics Engineers.
-  **ACM**. Member of the Association for Computing Machinery.
-  **ISAL**. Member of the International Society for Artificial Life.

Student Associations

- 2023 –  **UniNE**. President of the Computational Thinking Association (Algorithmia).
-  **UniNE**. Treasurer of the Association Suisse Alémanique & Friends (ASA&F).
- 2014 – 2016  **HES-SO**. Head of Communication of the Student Association (REH-SO).
- 2010 – 2012  **EPFL**. Webmaster of the Role-playing Association (JDRpoly).
-  **EPFL**. Committee Member of the Physics Students Association. (Irrotationnels)
-  **EPFL**. Member of the Robotics Association (RoboPoly).
- 2009 – 2010  **Northeastern University**. Member of the Society of Physics Students (SPS).

Skills

Languages	 French Native, English Fluent, Russian Proficient, German Basic
Code & Frameworks	 Python, JAX, C++, Pytorch, NetworkX, Scikit, Keras, JS, ...
Machine Learning	 Evolutionary, Deep learning, Edge Models, LLMs API & Local, ...
Computing	 Optimization, HPC, Graphs, Kubernetes, Distributed, Parallel, ...
Dev-Ops	 Git, Docker, Continuous Integration, IaaS, PaaS, SaaS, ...
Miscellaneous	 CAD, Electronics, 3D Printing, Knowledge Bases, $\LaTeX$, Blockchain, ...

Certifications

- 2013  **Entrepreneurship**. Granted by Venturelab, Switzerland.
- 2012  **Inventor Training** (CAD). Granted by Hurni Engineering, Switzerland.
- 2008  **Human Subjects Research**. Granted by CITI program, USA.

Major scientific achievements

Conference Proceedings

- [1] R. Claret, M. O'Neill, P. Cotofrei, and K. Stoffel, "Investigating hyperparameter optimization and transferability for es-hyperneat: A tpe approach," in *Proceedings of the Genetic and Evolutionary Computation Conference Companion*, 2024, pp. 1879–1887.
- [2] F. Karmali, K. Lim, A. Adatia, R. Claret, K. Nicoucar, and D. M. Merfeld, "Perceptual roll-tilt thresholds demonstrate visual-vestibular fusion," in *40th Annual meeting of Neuroscience, San Diego, CA, on November*, 2010, pp. 13–17.